

Randomized and low-rank approximation

Random column subsets of arbitrary fixed-sparsity matrices

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Rating rationale: Historical ratings for the original open target. Challenging because arbitrary support intersections defeat existing sparse-matrix estimates; community impact is understanding limits of sparse sketches and column selection.

Resolution: affirmative, 12 September 2026

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[Theorem 1.1](#) and [Corollary 1.2](#) of *Random column subsets of fixed-sparsity matrices: an unrestricted theorem* prove the full statement below. The finite exponential tail bound covers arbitrary signs, repeated columns and unrestricted support intersections. Setting the aspect-ratio lower bound to $1/(2C)$ yields the exact asymptotic target. No growing-sparsity rate is claimed.

A separate [independent Codex AI-agent audit](#) returned PASS without mathematical corrections, satisfying the **Solved** policy. AI assistance was used for review and submission preparation; that audit was informal automated review, not human peer review or formal verification. The later Lean verification is recorded [below](#). [Proof source](#) · [Submission record](#).

Huang, Rudelson and Tikhomirov retain conjecture and prior-result credit. The original target and ID remain unchanged; the earlier checks below are historical.

Lean proof and verification evidence

Lean verified, 29 September 2026. OpenAI Codex agents authored the [complete Lean project at immutable revision 4aa20f0e](#). The declaration `NLA.TR07.random_column_subsets : SolvesTR07` proves the entire original asymptotic target, including arbitrary signs, repeated column values and unrestricted support intersections. [Frozen definitions and target correspondence](#) use Euclidean singular values and the exact uniform subset law. The explicit subset-size validity condition and eventual sparsity account for the experiment's domain and irrelevant initial dimensions; eventual positive, diverging sample size is proved.

This submission ran a fresh unprivileged Ubuntu check in [run 36544197412](#). The [retained Linux receipt and logs](#) record successful Comparator statement matching, default-kernel replay, permitted-axiom checks and rejection controls for the immutable proof. The [transitive axiom report](#) lists only *propext*, *Classical.choice* and *Quot.sound*. [Two independent nonauthor AI-agent reviews](#) cover the full statement and proof; no human peer review or source-author endorsement is claimed.

The pins are Lean 4.33.1 and [Mathlib revision 0df444a3](#), with [pinned transitive dependencies](#). The [proof guide](#) and [formalization metadata](#) document regularization, reservoir reconstruction and the polynomial probability bound. The source's stronger exponential finite tail and positive-fraction singular-value theorem are not claimed as formalized. Mathematical-resolution credit remains Sidney Holden.

Reproduce on the [supported unprivileged Linux environment](#), from the repository root:

```
tools/lean/bootstrap.sh /tmp/nla-tr07-checker
tools/lean/verify.sh \
  randomized-and-low-rank-approximation/TR-07/lean \
  /tmp/nla-tr07-checker
```

Problem statement

Fix an integer $s \geq 2$ and $C \geq 1$. Suppose $r_k \leq k$, $k/r_k \rightarrow C$, and $n_k/r_k \rightarrow \infty$. For every deterministic sequence $M_k \in \{-1, 0, 1\}^{k \times n_k}$ with exactly s nonzero entries per column, and a uniformly random r_k -element column set I_k , prove or disprove

$$\forall \eta > 0, \quad \Pr\{\sigma_{\min}((M_k)_{:I_k}) > \eta\} \rightarrow 0.$$

Here $\sigma_{\min}(B) = \inf_{\|x\|_2=1} \|Bx\|_2$. There is no hypothesis controlling intersections of column supports.

Reference

Han Huang, Mark Rudelson, and Konstantin Tikhomirov, [Well-Invertible Column Subsets of Sparse Matrices Are Rare](#), §7, Conjecture 7.1; compare Theorem 1.3 for the additional structural hypothesis in the proved result.

Status check

Searches included “*unrestricted deterministic sparse sketches*” conjecture and the exact paper title with 2026. No subsequent resolution was located. The paper’s disproof of a particular SparseStack conjecture does not establish its own unrestricted deterministic conjecture.

Audit — 2026-09-10

Rechecked [Huang–Rudelson–Tikhomirov, Theorem 1.3 and Conjecture 7.1](#): a structural subclass is proved, while arbitrary deterministic supports remain conjectural. Title and sparse-subset follow-up searches found no full resolution. [Tikhomirov’s later embedding paper](#) studies random models and spectral-norm estimates, not all deterministic support patterns.